

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-1038
Project Name	Auto Ticket Classification & Routing System using Flow Designer
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Exact Keyword Dependency: Relies strictly on matching configured string values (e.g., 'WiFi', 'Projector').	Unconventional language, spelling mutations, or complex human descriptions will result in tickets remaining unclassified.	High
2	Multi-Issue Description Ambiguity: Descriptions referencing multiple criteria trigger only the first sequential execution rule pattern match.	Secondary technical problems inside multi-line logs go unseen, causing routing slips or poor assignment routing.	Medium
3	Monolingual Rule Execution Matrix: The Visual Studio Flow Designer logic stack evaluates text exclusively using English rules.	Inbound support strings written in regional dialects fail logic parameters entirely, requiring manual routing triage updates.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tuned primarily for internal school/college local testing profiles.	Deploy horizontal scaling architectures across regional server nodes to easily accommodate thousands of concurrently active users.
2	Data Storage	Relational storage schemas handled entirely inside local custom table instances (<code>`u_incident_workflow`</code>).	Introduce advanced archival indexing policies and data separation strategies to safely handle millions of high-volume log inputs.
3	Performance	Synchronous condition checks running structural lookups beneath 1 second.	Transition workflow structures into asynchronous job worker queues to safeguard database execution times during intense load peaks.
4	Security	Basic rule validation controls restricted via structural role configurations (<code>`u_incident_workflow_user`</code>).	Enforce end-to-end token validation profiles and deep contextual input filtering rules to actively protect against scripting injections.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate ServiceNow AI/ML Predictive Intelligence engines for Natural Language Processing (NLP) classification models.	Q1 2027	Eliminates keyword dependency traps by understanding synonyms, misspelled variations, and conversational text logs automatically.
Phase 3	Expand automated logic modules to cleanly map assignment groups and engineer routing paths by category.	Q2 2027	Drastically reduces manual handling overhead by passing tickets straight to specific support queues without tier-1 eyes.
Phase 4	Deploy interactive conversational logging workflows via omnichannel virtual agents (Teams/Portal chatbots).	Q4 2027	Empowers users to rapidly log structured operational tickets directly inside their daily communication hubs.